

Release numerics update

The ionization closure, the domain, the new face solver, and how much the top boundary matters

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- 1 Which ionization closure, and where can we trust it?
- 2 A new face solver: the SWMF exact Riemann problem.
- 3 A precision fix, briefly.
- 4 How much does the *top* boundary set the answer?

The release model

1.5D field-aligned single fluid

$$U = (\rho, \rho u, E)$$

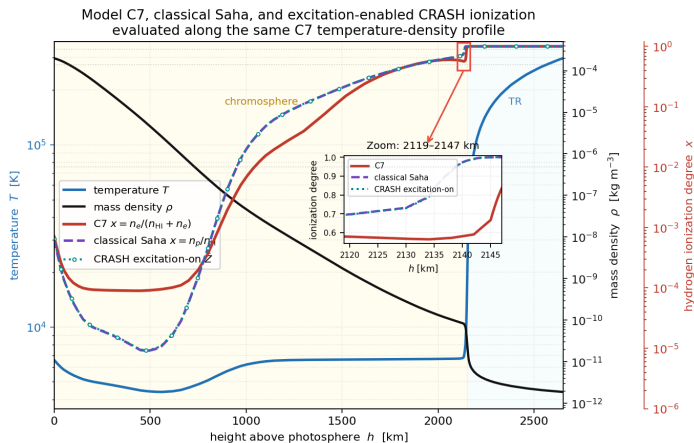
Saha equilibrium EOS

MUSCL–Hancock $\rightarrow$ Godunov

$\rightarrow$ implicit conduction

1600–2153 km, 661 cells

What sets hydrogen ionization in the chromosphere?



- Above ~ 1.8 Mm, Saha **tracks** the semi-empirical C7. Both cross $x = 0.5$ within 7 km.
- Lower down they part — and the **sign flips**: C7 is $10\times$ higher at 500 km, $3\times$ lower at 1200 km.
- Excitation physics does **not** close the gap.

C7 is non-LTE (Avrett & Loeser 2008); the other curves are LTE equilibrium on C7's own (T, n_H) .

Two consequences

We adopt Saha equilibrium

- One consistent, differentiable closure: x , n_e , p , e_{int} from (ρ, T) .
- x is **derived** — never stored, never advanced.
- Ionization energy is carried in e_{int} .

We truncate at 1600 km

- A mid-chromospheric reservoir — **not the photosphere**.
- Where we stop claiming the closure applies. Below is *outside the model*.
- Quasi-static, stratified, $v = 0$.

1600 km is a judgement call, not a derived height.

The release face solver: SWMF exact Riemann

Ported, not reinvented:

- `SWMF/util/CRASH/src/test_godunov.f90 :: get_godunov_flux`
- `SWMF/share/Library/src/ModExactRS.f90 :: exact_rs_pu_star, exact_rs_sample`

The Saha internal energy **splits exactly**:

$$e_{\text{int}} = \underbrace{\frac{p}{5/3 - 1}}_{\text{monatomic translation}} + \underbrace{e_{\text{ion}}}_{\text{rides in the offset } E_0}$$

Split $\rightarrow$ **solve** $\rightarrow$ **sample** $\rightarrow$ **carry** E_0

Split $\rightarrow$ solve $\rightarrow$ sample $\rightarrow$ carry

- 1 **Split.** Translational gas at $\gamma = 5/3$; ionization energy and gravitational potential into an offset $E_0 = (E - \frac{1}{2}\rho u^2 - p/(\gamma - 1))/\rho$.
- 2 **Solve** the local *ideal-gas* Riemann problem exactly.
- 3 **Sample** the self-similar solution at $x/t = 0$.
- 4 **Carry** E_0 with the contact, then rebuild the energy flux.

Why freezing composition is the right limit

Chromospheric hydrogen cannot re-equilibrate on a wave timescale: $\tau_{\text{ion}} \sim 10^5$ s in the mid-chromosphere, still $10\text{--}10^3$ s inside shocks — against a dynamical time of ~ 1 minute.

Carlsson & Stein 2002; Leenaarts+ 2007, 2020

Verification

- Toro's **five standard tests**, to the 6 published digits.
- Degenerate cases *exact*: a stationary contact carries identically zero mass.
- E_0 verified to advect passively — no effect beyond round-off.
- 4000s production run: **941 679 108 faces, all solved exactly, zero fallbacks.**

Release validation now fails on a nonzero fallback count.

The bulk EOS is still Saha equilibrium. Composition is frozen only *across the wave interaction* — this is not a finite-rate ionization model.

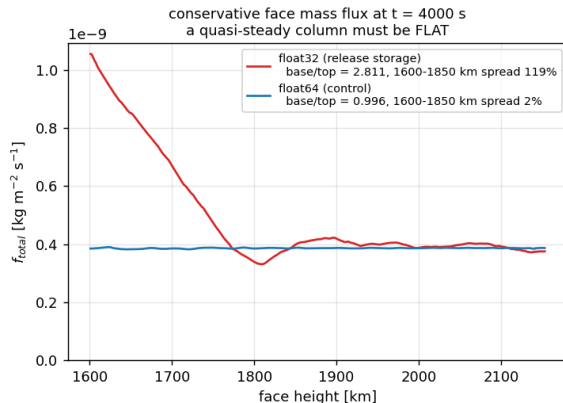
Aside: the lower-chromosphere mass flux was a storage bug

Continuity: a quasi-steady column must carry a **height-independent** mass flux. Ours did not.

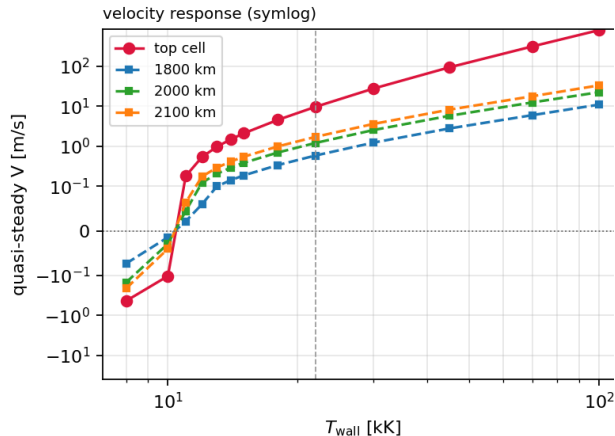
$$\rho \neq \Delta t \cdot \text{rhs}$$

In **619 of 660 cells** that increment is below half a float32 ULP of ρ — it rounds straight back. The density cannot respond.

A double-precision control, everything else identical, makes it flat. Not physics, not the solver, not the boundary. It is *not* cosmetic: top-of-domain evaporation drops $\sim 28\%$ (13.18 $\rightarrow$ 9.50 m/s), so every float32-era evaporation number was high.



How much does the top boundary set the answer?



One knob: T_{wall} at the 2153 km face.

8, 10, 11, 12, 13, 14, 15, 18,
22, 30, 45, 70, 100 kK

13 runs $\times$ 4000 s. Same C7 start, everything else fixed.

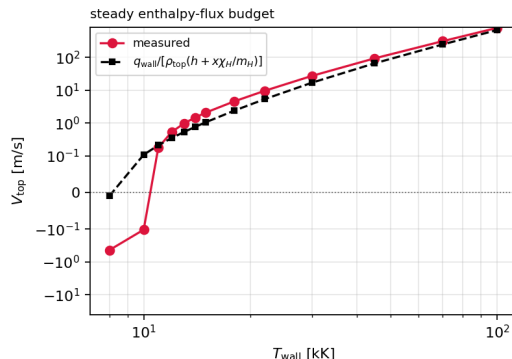
- $-0.4 \rightarrow +792$ m/s.
Four decades of V per decade of T_{wall} .
- Reverses near 10 kK: the column **drains**, not evaporates.
- Local slope 3.7 at our 22 kK.
10% in the wall $\Rightarrow$ 40% in the evaporation.

ρV flat to 14% in height: a mass-flux response. $V(h)$ rises only because ρ falls.

Conduction sets it; ionization pays for it

Three links, all measured:

- 1 **Thermostat.** T_{top} tracks T_{wall} to 0.3%, so $\Delta T \propto T_{\text{wall}}^{0.8}$ — self-limiting.
- 2 **Spitzer.** $\kappa \propto T^{2.50} \Rightarrow q_{\text{wall}} \propto T_{\text{wall}}^{3.3}$.
All the nonlinearity. $\text{Mach} \leq 0.015$.
- 3 **Export.** No radiative sink, so $V \simeq q_{\text{wall}} / [\rho_{\text{top}}(h + x\chi_H/m_H)]$.



Within a factor of two across four decades, no fitting. Ionization is 59% of that denominator at 22 kK.

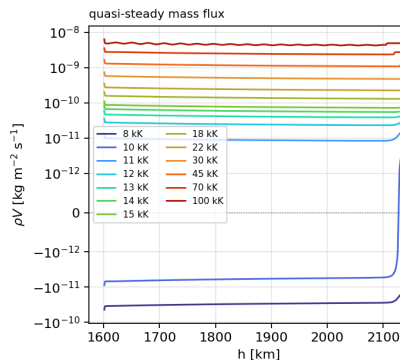
Status and open questions

Release: single-fluid Saha equilibrium mixture; MUSCL–Hancock + SWMF exact Riemann; implicit conduction; 107 test cases green.

- 1 Saha vs. finite relaxation.** Composition frozen *inside* the Riemann problem, re-equilibrated *after every step*.
Finite-rate network, or a relaxation source?
- 2 The top boundary is the dominant lever.** $V \propto T_{\text{wall}}^3$.
Couple to a real corona, or keep quoting a chosen wall?
- 3 Resolution.** N500 evaporation mass flux: not established as <10% grid-converged.
- 4 Minor, tracked.** One artifact at the first interior face — survives the precision control, so it is real.

Backup: what the wall sweep does *not* establish

- **Quasi-steady, not steady.** V_{top} still falling 2–6% per 500 s — magnitudes are upper bounds, the trend is not.
- **Reversal bracketed at 8–11 kK, not pinned.**
- **6 kK is outside the model:** the top cools below the Γ_1 table floor and aborts.
- **100 kK needed CFL 0.25.** Matched 70 kK control: 2.6×10^{-5} .
- **Back-pressure fixed** — only the thermal BC moves.



ρV flat with height: a real steady mass flux.

Backup: sources

Claim	Where it comes from
C7 vs Saha vs CRASH ionization	util/plot_c7_ionization_profile.py --mode compare
1600 km truncation	docs/doxygen/pages/physics_model.md; scenarios/model_column.cpp
Relaxation timescales	Carlsson & Stein 2002; Leenaarts et al. 2007; Leenaarts 2020
Toro tests, E_0 passivity, 0 fallbacks	tests/chromo_tests.cpp; docs/studies/numerics/swmf_godunov_flux_experiment.md
float32 vs float64 control	docs/studies/numerics/float32_precision_control_experiment.md; docs/studies/numerics/state_precision_release_cutover.md
Wall-temperature sweep	docs/studies/boundaries/upper_wall_temperature_sensitivity_recap.md; util/run_twall_sweep.sh, util/twall_sweep_diag.py
Figure provenance	figures/README.md; visualize_commands.md §14–16
Test counts	./build_omp/chromo_tests, current tree

Flux cutover: 022ea34 · Reference-free release: 34ecb00 · Precision cutover: 78109a9 · Wall sweep: 030347a